

Problem

Find $G(s)$ given that $g(t) = 2r(t) - 2r(t - 2)$.

Step-by-step solution

Step 1 of 3

Consider the following signal:

$$g(t) = 2r(t) - 2r(t - 2)$$

Here,

$$r(t) = \begin{cases} t & \text{for } t \geq 0 \\ 0 & \text{for } t < 0 \end{cases}$$

The Laplace transform of $g(t)$ is

$$L\{g(t)\} = L\{2r(t) - 2r(t - 2)\}$$

$$G(s) = L\{2r(t)\} - L\{2r(t - 2)\}$$

$$G(s) = 2L\{r(t)\} - 2L\{r(t - 2)\} \dots (1)$$

[Comments \(1\)](#)



Anonymous

how do I know what $r(t)$ equals at $t < 0$ and $t > 0$ if it isn't given in the problem statement?

Step 2 of 3

Refer to Table 15.1 for properties of the Laplace transform in the text book.

The Laplace transform of the $r(t)$ is,

$$L\{r(t)\} = L\{t\}$$

$$R(s) = \frac{1}{s^2}$$

Time shifting property of the Laplace transform is,

$$L[f(t - a)] = e^{-as} F(s)$$

Therefore,

$$L[r(t - t_0)] = e^{-st_0} R(s)$$

The Laplace transform of the $r(t - 2)$ is,

$$\begin{aligned} L\{r(t - 2)\} &= e^{-2s} \left(\frac{1}{s^2} \right) \\ &= \frac{e^{-2s}}{s^2} \end{aligned}$$

Step 3 of 3

Recall equation (1),

$$G(s) = 2L\{r(t)\} - 2L\{r(t-2)\}$$

Substitute $\frac{1}{s^2}$ for $R(s)$ and $\frac{e^{-2s}}{s^2}$ for $L\{r(t-2)\}$ in the equation.

$$\begin{aligned} G(s) &= 2\left(\frac{1}{s^2}\right) - 2\left(\frac{e^{-2s}}{s^2}\right) \\ &= \frac{2}{s^2}(1 - e^{-2s}) \end{aligned}$$

Thus, the Laplace transform of $g(t)$ is, $G(s) = \frac{2}{s^2}(1 - e^{-2s})$.